

AI AGENT KA PURA WORKING CODE LIKE A EXAMPLE



YE CODE KYA BANATA HAI?

Ye code ek **AI Agent chatbot** banata hai jo:

- User se terminal me question leta hai
- Samajhta hai ki:
 - Crypto price chahiye ?
 - Weather info chahiye ?
- Fir **khud decide karke function call karta hai**
- Real-time API se data lata hai
- Human-like answer deta hai
- Aur **previous conversation yaad rakhta hai** (multi-turn)

1 Imports – kya aur kyun?

```
import { GoogleGenAI, Type } from '@google/genai';  
import readlineSync from "readline-sync"  
import 'dotenv/config'
```

Matlab:

- **GoogleGenAI** → Gemini AI use karne ke liye
- **Type** → AI ko batane ke liye function me kis type ka data jayega
- **readline-sync** → Terminal se user input lene ke liye

- `dotenv/config` → `.env` file se API key lene ke liye
-

2 AI Client banana

```
const ai = new GoogleGenAI({});
```

- 👉 Ye AI ka **brain connect** karta hai
 - 👉 Yahin se Gemini model call hota hai
-

3 Tools (Real world ka kaam)

♦ Crypto function

```
async function cryptocurrency({coin}) { ... }
```

- 👉 Ye function **CoinGecko API** se crypto ka price nikalta hai
 - 👉 AI internet direct access nahi kar sakta
 - 👉 Isliye **tool/function** banana padta hai
-

♦ Weather function

```
async function weatherInformation({city}) { ... }
```

- 👉 Ye city ka **current weather** lata hai
-

4 ? Argument OBJECT me hi kyun pass kiya?

✗ Galat

```
cryptocurrency("bitcoin")
```

AI confuse ho sakta hai:

- Ye kya hai?
- Coin hai ya city?

✅ Sahi

```
cryptoCurrency({ coin: "bitcoin" })
```

Fayda:

- Clear hota hai kaunsa data kya hai
- Aage extra cheez add kar sakte ho

```
{ coin:"bitcoin", currency:"inr" }
```

👉 Isliye **GenAI tools hamesha object lete hain**

5 Function declaration (AI ko rules batana)

```
const cryptoInfo = {  
  name:"cryptoCurrency",  
  description:"Crypto info deta hai",  
  parameters:{ ... }  
}
```

👉 Ye AI ko batata hai:

- Function ka naam
- Kab use karna hai
- Kaunsa argument mandatory hai

Same cheez weather ke liye bhi

6 Tools AI ko dena

```
const tools = [{  
  functionDeclarations:[cryptoInfo, weatherInfo]  
}];
```

👉 AI ko bola:

Tumhare paas 2 powers hain
Crypto aur Weather

7 Tool name ko real function se jodna

```
const toolFunctions = {  
  cryptocurrency,  
  weatherInformation  
}
```

👉 AI sirf **function ka naam** boleگا

👉 JavaScript ko pata hona chahiye:

Is naam par kaunsa function run hoga

8 History = Memory 🧠

```
const History = [];
```

👉 AI khud kuch yaad nahi rakhta

👉 Is array me:

- User ke questions
- AI ke answers
store hote hain

👉 Isi se **multi-turn conversation** hoti hai

9 runAgent() – asli game yahin hai

```
ai.models.generateContent({  
  model:"gemini-2.5-flash",  
  contents:History,  
  config:{ tools }  
})
```

👉 AI:

- Puri history padhta hai
- Decide karta hai:
 - Text reply?
 - Ya function call?

♦ Agar function call hua

```
result.functionCalls.length > 0
```

Flow:

1 AI bola:

👉 `weatherInformation({city:"Delhi"})`

2 JS ne function run kiya

3 API se data aaya

4 Data wapas AI ko diya

5 AI ne final answer banaya

♦ Agar normal text aaya

```
console.log(result.text)
```

👉 Seedha user ko reply

10 Terminal loop (chat chalane ke liye)

```
while(true){  
  const question = readlineSync.question(...)  
}
```

👉 Jab tak user `exit` nahi likhe

👉 Chat chalta rahe

PURE CODE KA FLOW (MOST IMPORTANT)

User question

↓

History me save

↓

Gemini AI ko bheja

↓

AI decide karta:

├ Function call? → API → data → AI → answer

└ Direct text answer

↓

Answer print

↓

Next question

Short me samajh lo

Cheez

Matlab

Tool	Real world ka kaam
Object args	Clear & future safe
History	Memory
Function call	AI ka decision
Agent	AI + tools + logic